

OUR OBJECTIVES

FORM PRESENTATION

1

Dataset overview

1

Dataset overview

The purpose of this dataset is to provide an initial overview prior to conducting a more in-depth analysis, and to supply information that reflects the actual situation.

2

**Statistical
methods used**

2

Statistical methods used

In the analysis, we employed several statistical methods to gain a clearer understanding of the data and to extract meaningful insights.

3

Main findings

3

Main findings

The dataset analysis revealed key variations, correlations, and some outliers, highlighting areas for improvement and opportunities to enhance performance.

4

Conclusions and recommendations

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Conclusions and recommendations

Focus on key variables, address outliers, and use positive trends to improve performance.

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WELCOME EVERYONE

Dataset overview • Statistical methods used • Main findings
Conclusions and recommendations

Dataset overview

Product_ID	Production_Line	Shift	Operator_ID	Machine_ID	Date
PROD_00001	2	B	OP_004	MACH_01	5/22/2020
PROD_00002	5	B	OP_018	MACH_03	3/1/2017
PROD_00003	4	C	OP_005	MACH_05	7/23/2023
PROD_00004	3	A	OP_016	MACH_06	4/18/2021
PROD_00005	1	C	OP_014	MACH_03	8/15/2014
PROD_00006	1	A	OP_019	MACH_04	7/29/2019
PROD_00007	1	C	OP_014	MACH_05	3/3/2019
PROD_00008	5	C	OP_009	MACH_03	12/2/2011
PROD_00009	3	A	OP_006	MACH_04	5/16/2021
PROD_00010	4	B	OP_017	MACH_05	5/13/2014

Statistical methods used • Main findings
Conclusions and recommendations

Dataset overview

Temperature_C	Pressure_psi	Speed_rpm	Production_Time_min	Quality_Score	Defect_Flag
221.3	215.4	1221.8	124.6	83.67992269	Bad
221.2	207.2	1240.7	116.5	75.92869879	Bad
220.8	199.5	1162.9	128.6	81.97949183	Bad
220.2	219.4	1238	122.6	80.94228253	Bad
219.7	198.2	1158.2	130.2	85.03999371	Bad
218.9	218.3	1204.6	127.8	82.88220961	Bad
217.9	185.8	1162.7	130.2	82.43999371	Bad
217.6	207.5	1128.2	130.1	81.73420518	Bad
217.5	205.1	1218.4	124.7	83.16624287	Bad
217.4	200.6	1158	128.4	85.28220961	Good

Statistical methods used • Main findings
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Statistical methods used

Mean (Average):

We used the mean because it is the foundation for calculating standard deviation, variance, and other operations. It is useful for knowing the typical value in the dataset and understanding the general trend (whether most values are high, medium, or low).

Median:

We used the median because it shows the middle value after arranging the data. It helps, together with the mean, to see if the distribution leans left, right, or is symmetric, giving a clear view of the dataset.

Standard Deviation:

This measures how much the data is spread out. We used standard deviation instead of variance because it gives a more accurate measure of data dispersion, whereas variance is the square of the spread.

Statistical methods used

Variance:

Variance measures how spread out the data is, but since it uses squared values, it is less intuitive compared to standard deviation.

Minimum Value:

This is the smallest value in the dataset. It is important for understanding the data range and is useful in decision-making and analysis.

Maximum Value:

This is the largest value in the dataset. It is also important for understanding the data range and helps indicate the highest value reached in the dataset.

Range:

The range is calculated by subtracting the smallest value from the largest value. It helps to understand the spread of the data and allows for comparisons.

Statistical methods used

Mode:

We calculated the mode and the percentage of "good" and "bad" in the defect flag to check whether we are on the right track or not.

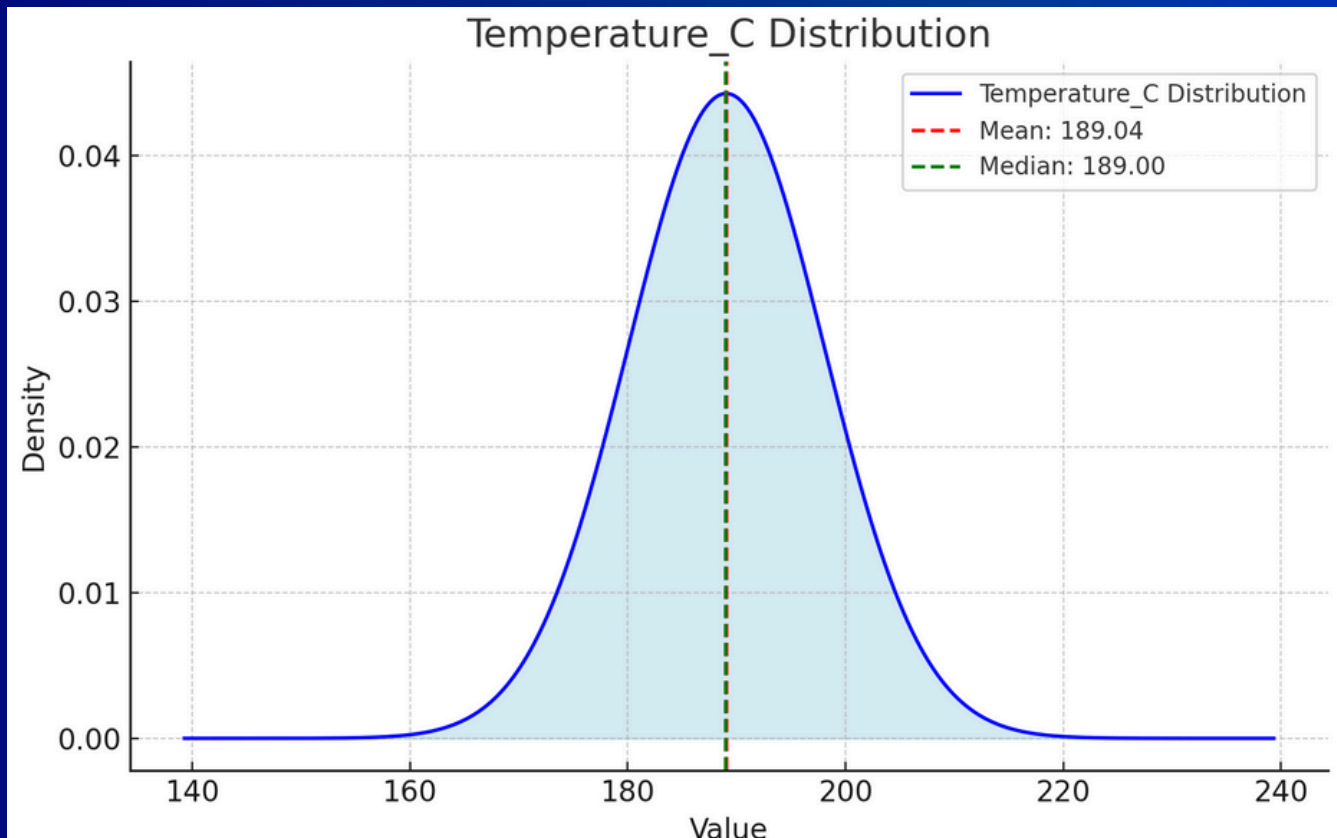
Interquartile Range (IQR):

The IQR measures the true spread of the data without being affected by outliers, because it focuses only on the middle portion of the dataset. It gives a more accurate picture of the range where most values actually lie.

Statistical methods used

Variable	Mean	Median	Standard deviation	Minimum value	Maximum Value
Quality	83.37	83.36	3.98	68.84	97.84
Temperature	189.04	189.00	9.01	157.30	221.30
Pressure	201.46	201.40	12.28	159.10	243.90
Speed	1214.40	1213.75	50.21	1058.60	1368.50
Production Time Minimum	123.70	123.70	5.47	106.90	141.60

Main findings



Mean
189

Measures of Central Tendency

Median
189

Measures of Central Tendency

Std
9

Measures of Dispersion

Min
157.3

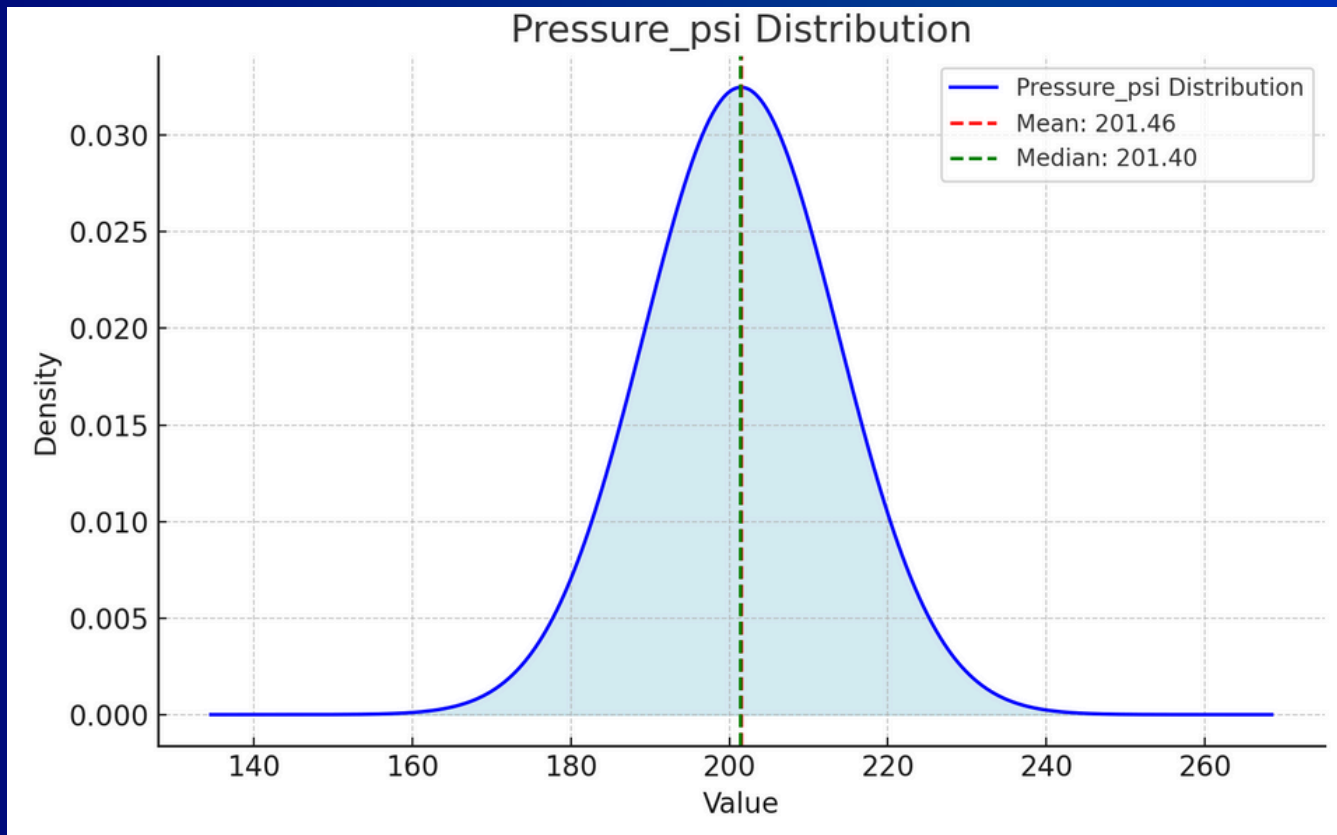
Measures of Dispersion

Max
221.3

Measures of Dispersion

Conclusions and recommendations

Main findings



Mean
201.4

Measures of Central Tendency

Median
201.4

Measures of Central Tendency

Std
12.28

Measures of Dispersion

Min
159.1

Measures of Dispersion

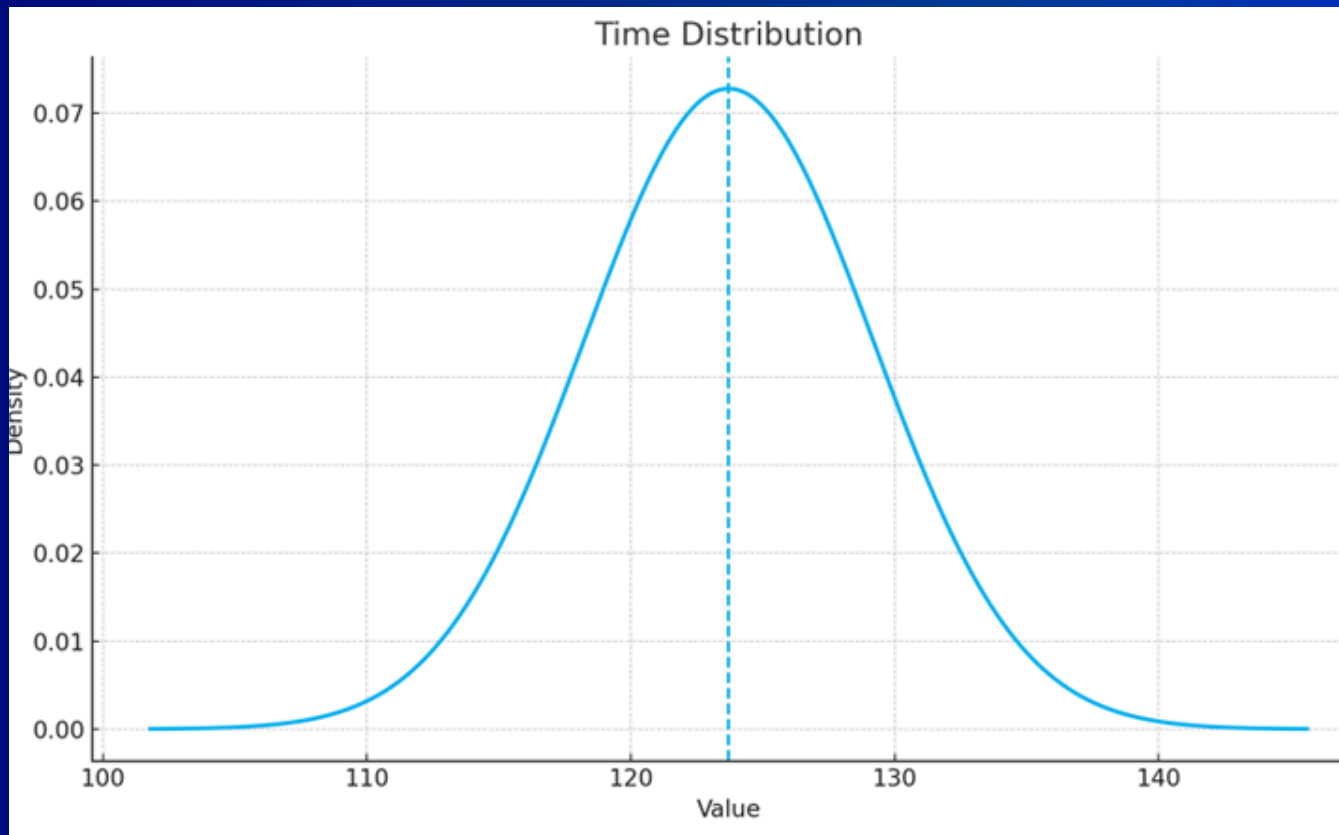
Max
243.9

Measures of Dispersion

Conclusions and recommendations

GROUP C

Main findings



Mean
123.7

Measures of Central Tendency

Median
123.7

Measures of Central Tendency

Std
5.47

Measures of Dispersion

Min
106.9

Measures of Dispersion

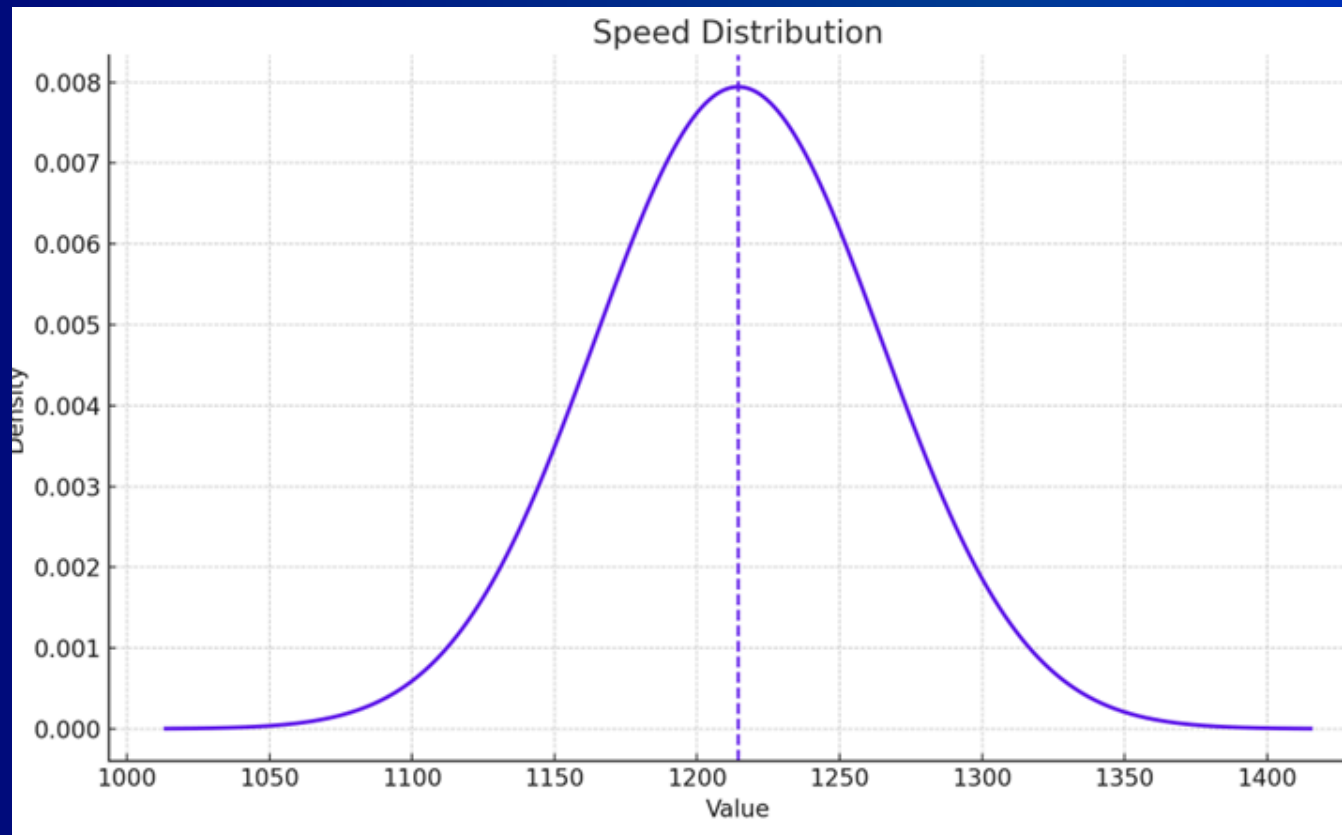
Max
141.6

Measures of Dispersion

Conclusions and recommendations

GROUP C

Main findings



Mean
1214

Measures of Central Tendency

Median
1214

Measures of Central Tendency

Std
50.21

Measures of Dispersion

Min
1058

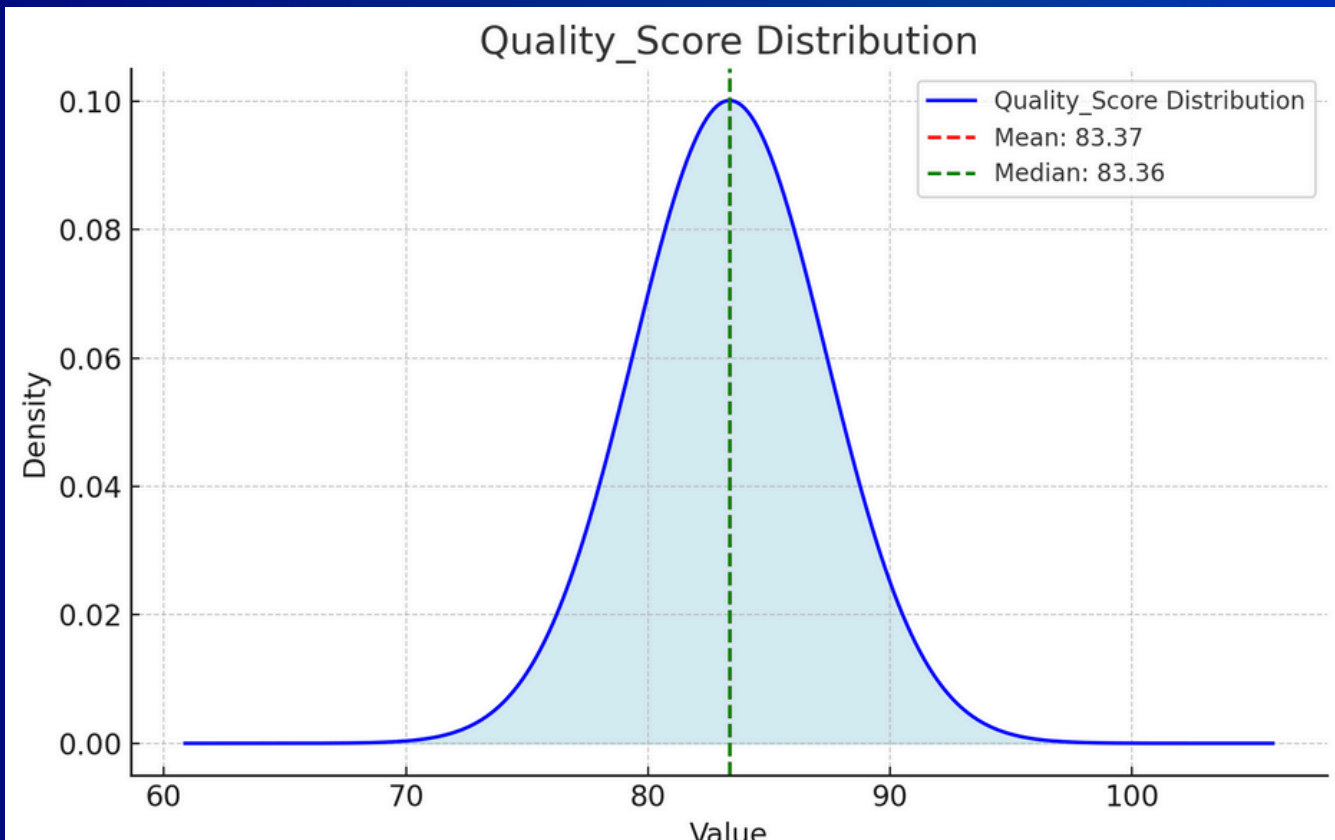
Measures of Dispersion

Max
1368

Measures of Dispersion

Conclusions and recommendations

Main findings



Mean
83.3

Measures of Central Tendency

Median
83.3

Measures of Central Tendency

Std
3.9

Measures of Dispersion

Min
68.8

Measures of Dispersion

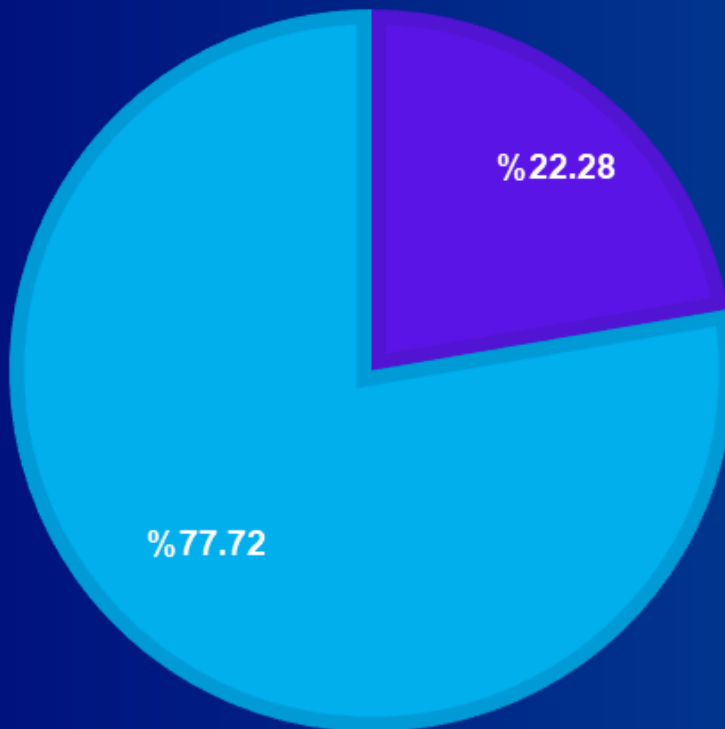
Max
97.8

Measures of Dispersion

Conclusions and recommendations

Main findings

Descriptive Statistics by Defect Flag:



Bad

Good

MODE
3886

The results indicate that the vast majority of the products are classified as "Bad" (defective).

MODE
1114

The results indicate that good products do not exceed.

Conclusions and recommendations

Main findings

Quality Score - Descriptive Statistics by Shift:

Mean 84.5	Shift A	Mean 83	Shift B	Mean 82.41	Shift C
Median 84.5	Shift A	Median 82.9	Shift B	Median 82.3	Shift C
Var 14.8	Shift A	Var 14.7	Shift B	Var 15.95	Shift C
Std 3.8	Shift A	Std 3.8	Shift B	Std 4	Shift C

Conclusions and recommendations

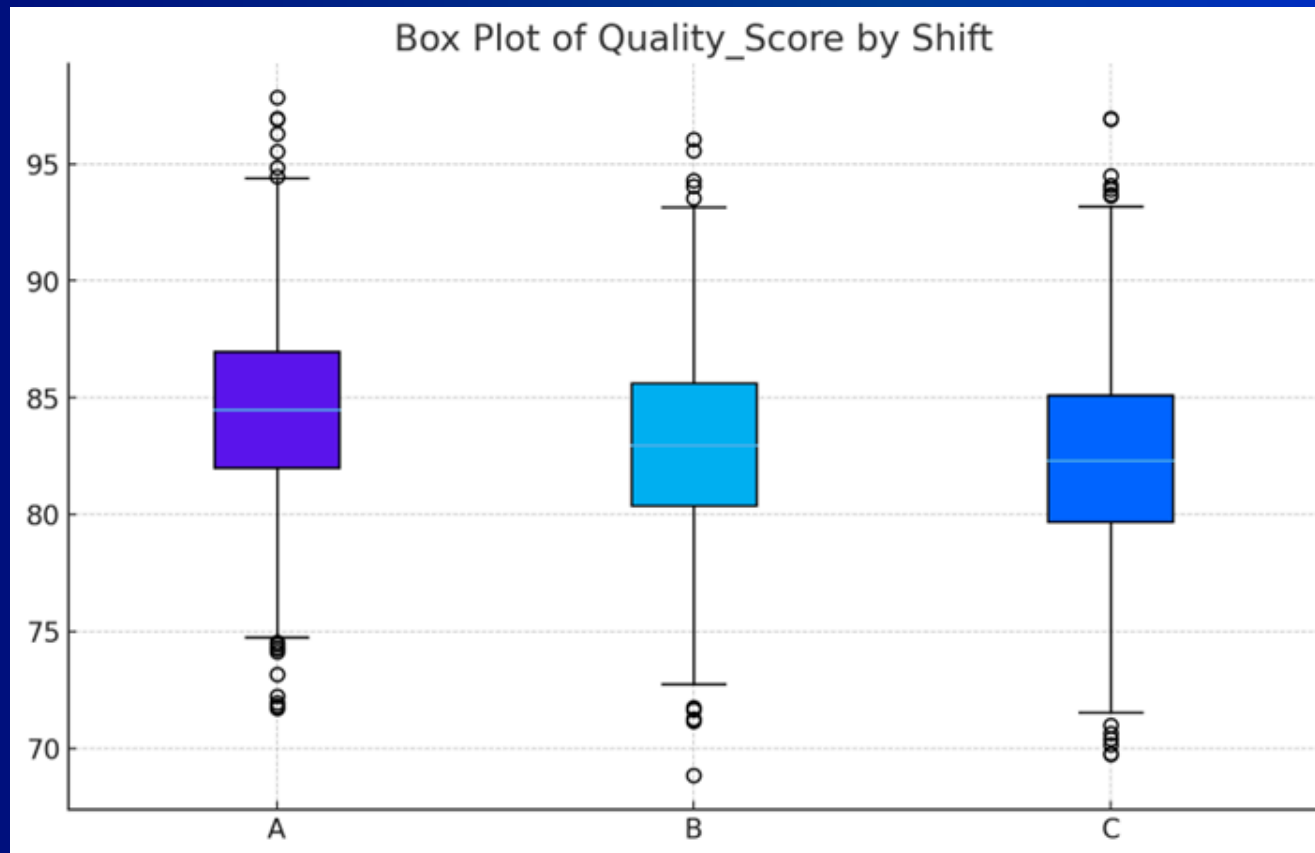
Main findings

Quality Score - Descriptive Statistics by Shift:

Range 29	Shift A	Range 28.96	Shift B	Range 28.96	Shift C
IQR 5.28	Shift A	IQR 5.28	Shift B	IQR 5.28	Shift C
Percentage of Bad 79%	Shift A	Percentage of Bad 78%	Shift B	Percentage of Bad 76%	Shift C
Number of Bad 1403	Shift A	Number of Bad 1405	Shift B	Number of Bad 1078	Shift C

Conclusions and recommendations

Main findings



Conclusions and recommendations

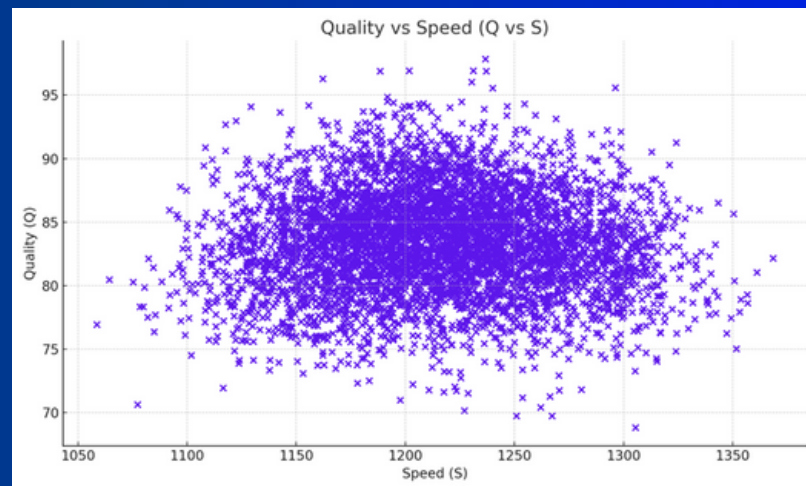
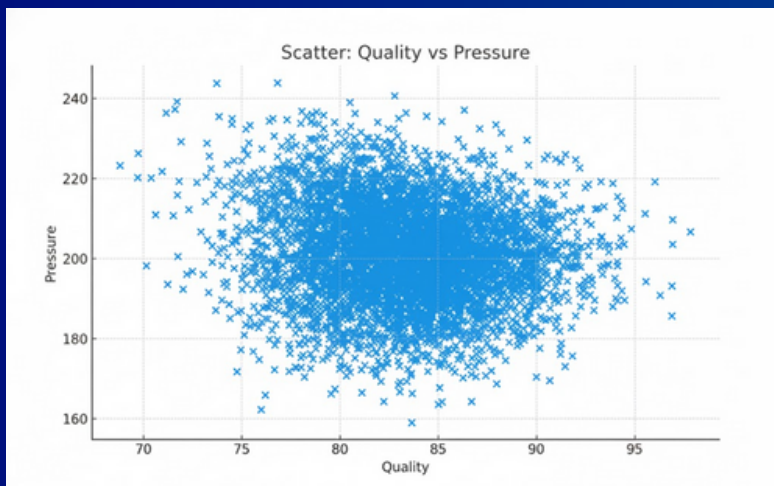
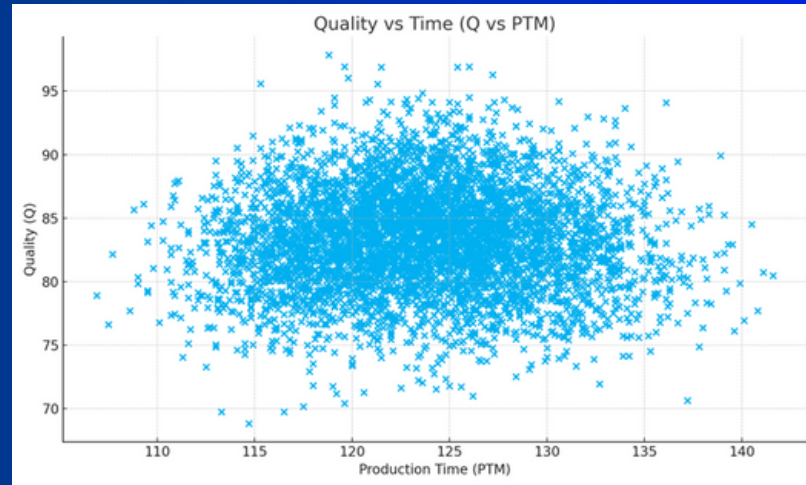
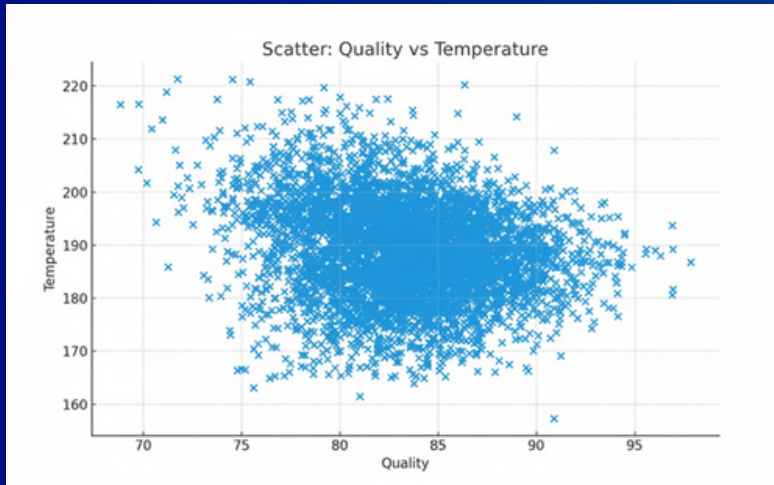
Main findings

Correlation Coefficients:

DATA	Quality	Temperature	Pressure	PTM	Speed
Quality	1.0	-0.213	-0.13	0.007	-0.013
Temperature	-0.213	1.0	0.584	-0.016	0.015
Pressure	-0.13	0.584	1.0	-0.021	0.02
PTM	0.007	-0.016	-0.021	1.0	-0.933
Speed	-0.013	0.015	0.020	-0.933	1.0

Conclusions and recommendations

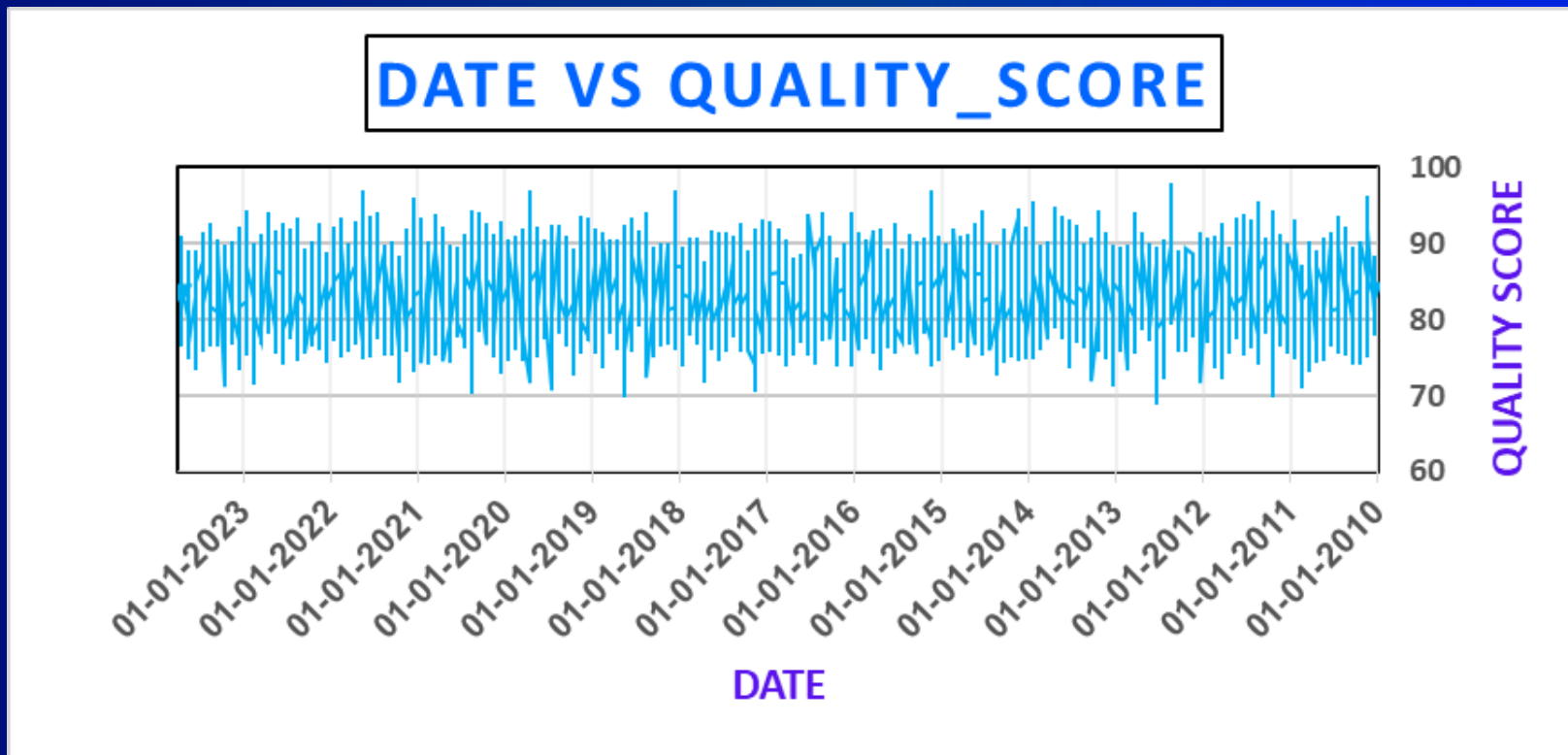
Main findings



Conclusions and recommendations

Main findings

Time Series plot:



Conclusions and recommendations

Conclusions and recommendations

After going through the dataset and analyzing everything in detail, we ended up with several important points that give us a clear picture of what's happening in the production process. First of all, the quality level is mostly stable overall, and it doesn't really change much over the years. This means the process is running at the same level without any real improvement or decline. However, we did notice a lot of day-to-day fluctuations in the quality scores, which shows that the process is affected by different factors that change from day to day, causing a lot of "noise" in the output. We also found that quality has no strong linear relationship with temperature, pressure, speed, or production time. So within the current operating ranges, none of these variables are directly increasing or decreasing the quality. One of the strongest findings was the very strong negative correlation between speed and production time. The faster the speed, the shorter the production time — which makes perfect sense. But the biggest issue we discovered was the extremely high defect rate. Across all shifts, more than 75% of the products are labeled as "Bad," which means the problem isn't coming from a specific team or shift — it's most likely a process-level issue or something related to how quality is being measured.

Conclusions and recommendations

The time-series analysis also showed that quality doesn't get better or worse over the years. It stays at the same level from 2010 to 2023, just with constant fluctuations. In short, the process is running, but it's not really improving.

Recommendations:

- Reevaluate the entire production process, because the defect rate is extremely high and needs to be understood and fixed.
- Work on reducing the daily fluctuations in quality, since the high volatility suggests that some factors are not being controlled properly.
- Monitor temperature and pressure more closely, as they have a moderate relationship and could impact stability if they go out of range.
- Use the strong relationship between speed and production time to optimize workflow and improve efficiency without hurting quality.
- Apply continuous monitoring and regular statistical analysis to identify issues early instead of waiting for them to pile up.
- Review the quality classification criteria, because having such a high percentage of "Bad" products may mean the evaluation method needs adjustment or clarification.

GROUP C

ANY QUESTIONS ?

GROUP C

THANK YOU !!

FOR LISTENING

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